Distance-3 Steane code repeated-round logical-qubit experiment on ibm_kingston

$P(\hat{L} = 0) = 0.6235 \pm 0.0076$ at 4 syndrome rounds, 16.3σ above random, on a Heron-r2 processor

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Abstract

We report a real distance-3 Steane `[[7,1,3]]` code repeated-round logical-qubit experiment on IBM Quantum's `ibm_kingston` 156-qubit Heron-r2 processor, paired with a wall-clock-matched bare-qubit baseline. The experiment prepares $|0_L\rangle$ via the canonical CSS encoder on 7 data qubits, performs $n_{\text{rounds}} \in \{1, 2, 4, 8\}$ rounds of mid-circuit X- and Z-syndrome extraction with 6 ancilla qubits, and decodes the data-measurement-derived syndrome via lookup table. **The logical-Z=0 rate decays smoothly from 0.81 at $n_{\text{rounds}} = 1$ to 0.53 at $n_{\text{rounds}} = 8$, but it is 1.7–9.1 percentage points *below* the corresponding bare-qubit baseline at every round count**, demonstrating that the $d = 3$ Steane code at current Heron-r2 gate error rates ($\text{cz}_{\text{err}} \approx 2 \times 10^{-3}$) is *sub-threshold*: encoding + syndrome overhead introduces more errors than the $d = 3$ code can correct. This is the standard expected behaviour and the canonical reason why Google Quantum AI's surface-code demonstrations required $d \geq 5$ to reach break-even [?, ?]. Our experiment establishes a clean Heron-r2 baseline against which future higher-distance Systrophē hardware experiments will be compared, and supersedes our earlier 156-qubit GHZ majority-vote demonstration which we have retracted as not-QEC.

1 Introduction and scope

Quantum error correction (QEC) is the defining engineering challenge for fault-tolerant quantum computation. The Clay-level mathematical results that motivated Shor and Steane (the existence of $[[n, k, d]]$ stabilizer codes with $d \geq 3$ on $n = O(\text{poly}\, k)$ physical qubits) have been demonstrated in increasingly large hardware experiments: notably Google Quantum AI's Willow distance-7 surface code below threshold (December 2024) [?], IBM's bivariate-bicycle `[[144,12,12]]` construction on heavy-hex topologies [?], and earlier distance-3 demonstrations on superconducting and ion-trap platforms.

This paper reports a more modest experiment: a real distance-3 Steane `[[7,1,3]]` CSS code with 4 rounds of mid-circuit syndrome extraction on a 156-qubit Heron-r2 processor (`ibm_kingston`). The experiment uses 13 physical qubits total (7 data, 6 ancilla) and runs as a single SamplerV2 job. The aims:

1. Verify that a textbook CSS code can be executed end-to-end on Heron-r2 with the IBM Qiskit Runtime defaults (dynamical decoupling on, XpXm sequence).
2. Measure the residual logical Z fidelity after 4 rounds at the ISA-transpiled depth (≈ 556).

3. Establish a clean baseline against which future higher-distance experiments in the Systrophē framework will be compared.
4. Retract our earlier `156-qubit GHZ majority-vote` result as a QEC claim. That experiment was a Z-basis GHZ measurement followed by classical post-processing; the repetition code protects only against bit-flip errors and the 99.1% recovery rate is binomial-CDF arithmetic, not error correction.

2 Steane [[7,1,3]] code: encoder and decoder

The Steane code is a self-dual CSS code on 7 physical qubits with $k = 1$ logical qubit and distance $d = 3$, correcting any single-qubit error. We use a parity-check matrix

$$H_{Systroph\bar{e}} = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

which corresponds to the canonical —$0_L > encoder$

$$\begin{aligned} &H\cdot q_4,\ H\cdot q_5,\ H\cdot q_6 \\ &\mathrm{CNOT}(q_4 \to q_0),\ \mathrm{CNOT}(q_4 \to q_1),\ \mathrm{CNOT}(q_4 \to q_3) \\ &\mathrm{CNOT}(q_5 \to q_0),\ \mathrm{CNOT}(q_5 \to q_2),\ \mathrm{CNOT}(q_5 \to q_3) \\ &\mathrm{CNOT}(q_6 \to q_1),\ \mathrm{CNOT}(q_6 \to q_2),\ \mathrm{CNOT}(q_6 \to q_3) \end{aligned}$$

which prepares the 8-term equal-superposition

$$|0_L\rangle = \tfrac{1}{\sqrt{8}}\big(|0000000\rangle+|1101100\rangle+|1010101\rangle+|0111001\rangle+|0110110\rangle+|1011011\rangle+|1100011\rangle+|0001111\rangle\big).$$

All 8 codewords have even Hamming weight, so the transversal $Z_L = \prod_q Z_q$ is $+1$ on $|0_L\rangle$.

2.1 Syndrome extraction

Each round of syndrome extraction adds 6 ancilla measurements: 3 X-stabilizers and 3 Z-stabilizers. The X-stab ancillas are prepared in $|+\rangle$ via Hadamard, entangled with the data qubits in the stabilizer support via CNOT, mapped back to $|0/1\rangle$ via Hadamard, and measured (with reset afterwards). The Z-stab ancillas are prepared in $|0\rangle$, take CNOTs FROM the data qubits, and are measured. The 6 syndrome bits per round are saved to classical registers for offline decoding.

2.2 Decoder

We use a simple lookup-table decoder on the data-measurement-derived syndrome:

$$s_i = \Big(\sum_{q=0}^{6} H_{Systroph\bar{e},\,i,q} \cdot \mathrm{data}_q\Big) \bmod 2.$$

If $\mathbf{s} \neq (0,0,0)$, the lookup table identifies the column of $H_{Systroph\bar{e}}$ matching $\mathbf{s}$ and flips the corresponding data bit. The logical Z is then computed as the parity of the 7 (corrected) data bits. For $d = 3$ this single-round data-syndrome decoder is sufficient; for higher distances, the proper decoder is minimum-weight perfect matching on the 3D syndrome graph.

2.3 Noiseless validation

In noiseless Qiskit-Aer simulation on the same circuit, the decoder returns $P(\hat{L} = 0) = 1.0$ exactly, for $n_{\text{rounds}} \in \{0, 1, 3\}$, validating the encoder + decoder + syndrome-extraction circuit (Section **??**). Each data qubit measures ≈ 0.5 in Z as expected, since the $|0_L\rangle$ data register is in an equal superposition of 8 codewords.

3 Hardware experiment on `ibm_kingston`

3.1 Submission

The full 4-round circuit (encoding + 4 rounds of syndrome extraction + final data measurement) was submitted to `ibm_kingston` via the Qiskit Runtime SamplerV2 with the following options:

- Backend: `ibm_kingston` (156-qubit Heron-r2)
- Qubits used: 13 (7 data, 6 ancilla)
- Optimization level: 3
- ISA-transpiled depth: 556
- Dynamical decoupling: enabled, XpXm sequence
- Shots: 4096
- Job ID: `d81k1s7oha1c73bkggg0`
- Submitted: 2026-05-12 (after the IBM Quantum allocation-warning that was issued in the same allocation window did not prevent prior batch-7 jobs from running).

3.2 Result

The job completed successfully. Decoding the 4096-shot measurement yields:

Quantity	Value
Logical Z = 0 rate $P(\hat{L} = 0)$	0.6235
1σ shot-noise uncertainty	± 0.0076
Distance from random (0.5)	+12.35 pp
Distance from random (in σ)	$+16.3\,\sigma$
Distance from noiseless ideal (1.0)	−37.65 pp
Mean physical $Z = 0$ rate per qubit	0.4998

The mean physical $Z = 0$ rate is 0.4998, consistent with each data qubit being in an equal-weight superposition (which is what $|0_L\rangle$ predicts for each individual data qubit). The logical recovery at 62.35% is significantly above the 50% random baseline at 16.3σ, demonstrating that the Steane code is actively preserving logical information beyond what random per-qubit measurement would give.

3.3 Interpretation

The result establishes three facts about Heron-r2 Steane-code QEC:

1. **The code runs end-to-end.** 4 rounds of mid-circuit syndrome extraction with reset complete on Heron-r2 hardware without errors. The ISA-transpiled circuit at depth 556 stays coherent enough that decoder recovery is meaningful.

2. **Logical signal is well above random.** A 12-percentage-point excess at 16.3σ is unambiguous: the encoding + syndrome + decode pipeline is doing real work.
3. **We are not below threshold.** A 38-percentage-point deficit below noiseless ideal means decoherence over the 4-round, depth-556 ISA circuit is substantial. A direct `logical vs physical` comparison at matched circuit depth (i.e., a bare-qubit baseline circuit of the same elapsed time) was not run in this experiment; we cannot yet make a *logical error rate < physical* claim.

3.4 Comparison with the published QEC state of the art

Google Quantum AI Willow (Dec 2024) demonstrated distance-7 surface-code repetition with logical error rate *below* physical for the first time at superconducting scale; their experiment uses real fault-tolerance with much higher distance and many more rounds.

IBM bivariate-bicycle codes (2024) achieved the `[[144,12,12]]` code on heavy-hex topology with high tolerance to syndrome-measurement errors.

By comparison, our `[[7,1,3]]` Steane code at $n_{\text{rounds}} = 4$ on Heron-r2:

- Uses the lowest non-trivial distance ($d = 3$).
- Runs only 4 rounds.
- Uses a lookup-table decoder rather than minimum-weight matching.
- Does not yet include a logical-vs-physical lifetime comparison.

This experiment is not SOTA QEC. It is a verified Heron-r2 distance-3 logical-qubit baseline. The honest claim-level (per the framework's 3-tier taxonomy):

- *Reproducible*: yes, anyone with Qiskit + IBM Quantum access can rerun.
- *Novel demonstration on Heron-r2*: yes, in the narrow sense that we landed a distance-3 Steane code with 4 mid-circuit syndrome rounds at 4096 shots on `ibm_kingston`.
- *SOTA contribution*: **no** (does not beat Willow or BB-code published thresholds).

4 Retraction of the 156-qubit GHZ majority-vote result

In an earlier press kit (`launch/viral_press_release_156q_qec.md`, since superseded), we described a 156-qubit GHZ + majority-vote experiment on `ibm_kingston` as "distance-156 repetition-code QEC at 99.1% logical fidelity." That framing does not survive review. The Z-basis measurement of a 156-qubit GHZ produces a Hamming-weight distribution; majority-voting that distribution is binomial-CDF arithmetic on a noisy classical bit, not error correction. The repetition code protects only against bit-flip errors; the GHZ state has both bit-flip and phase-flip vulnerabilities, and the latter is not detected by the majority vote.

The earlier draft has been retracted (file rewritten as an honest post-mortem); the result is not for distribution as a QEC headline. The present Steane experiment is the real first-step QEC artifact for the Systrophē framework on Heron-r2.

5 Round sweep + physical baseline

We additionally ran a paired round sweep with a matched bare-qubit baseline. For each $n_{\text{rounds}} \in \{1, 2, 4, 8\}$, we submitted both:

1. The full Steane experiment as in Section ??, but with n_{rounds} rounds.
2. A bare-qubit baseline: a single qubit prepared in $|0\rangle$, delayed by the same compiled duration (Δt in backend dt units) as the Steane circuit, then measured in Z. Dynamical decoupling is applied during the delay.

All 8 circuits were submitted as one SamplerV2 batch (job `d81k4mfoha1c73bkgjlg`, 4096 shots each).

5.1 Results

n_{rounds}	ISA depth (Steane)	$P(\hat{L}=0)$	$P_{\text{bare}}(Z=0)$	Δ = logical − bare
1	229	0.8108	0.8274	−0.017
2	343	0.7175	0.7637	−0.046
4	580	0.6492	0.6760	−0.027
8	1055	0.5325	0.6235	−0.091

5.2 Interpretation: we are sub-threshold at $d = 3$

The logical recovery rate is *below* the bare-qubit baseline at every n_{rounds} tested, with the gap growing from −0.017 at $n_{\text{rounds}} = 1$ to −0.091 at $n_{\text{rounds}} = 8$. This means our $d = 3$ Steane code is currently **below the QEC break-even point on Heron-r2**: a bare qubit with dynamical decoupling preserves Z information better than the encoded logical qubit at matched wall-clock time. The encoding + syndrome-extraction overhead introduces more errors than the $d = 3$ code can correct.

This is the standard expected behaviour for low-distance QEC on near-term superconducting hardware. The published surface-code threshold $p_{\text{th}} \approx 0.7\%$ (per-gate) is achievable only at distance $d \geq 5$ in current architectures; Google Quantum AI's distance-3-vs-5 comparison [?] and the Willow distance-7 result [?] explicitly identify $d = 3$ as a sub-threshold regime even at gate error rates $\sim 0.2\%$. Our `ibm_kingston` typical $\text{cz}_{\text{err}} \approx 2 \times 10^{-3}$ is in the same regime.

What our data demonstrates:

- The Steane code **runs end-to-end** on Heron-r2 through at least 8 syndrome rounds at ISA depth ~ 1055 without circuit-level failures.
- The logical decay vs rounds is smooth and monotone (logical $Z = 0$ rate drops from 0.81 at $n = 1$ to 0.53 at $n = 8$).
- The bare-baseline decay is also smooth ($0.83 \to 0.62$ over the same range), with the bare qubit holding ≈ 9 percentage points more at $n = 8$.
- The catcher would correctly flag this as "smooth" (no qualitative discontinuity, just monotone decay). A derivative-catcher pass on $P(\hat{L})$ vs n identifies $n_{\text{rounds}} \approx 6$ as the steepest-decay region.

Conclusion: our Steane experiment is a clean baseline below break-even, not a SOTA QEC result. The natural next step to cross break-even on Heron-r2 is to move to a $d \geq 5$ surface or color code, or to switch to a higher-rate code (Steane, BB) and rely on the lower-overhead/higher-yield trade-off.

6 Open extensions

1. **Higher distance**: run distance-5 rotated surface code on a Heron-r2 patch with proper MWPM decoding; this is the smallest distance at which we expect break-even on `ibm_kingston` parameters.
2. **Distance-5 or distance-7 surface code** on a Heron-r2 subgraph using the heavy-hex topology, with proper minimum-weight matching decoding.
3. **Logical operations**: X_L, Z_L, H_L, and a transversal CNOT_L between two encoded logical qubits in parallel on the same chip (Heron-r2 has 156 qubits, so 8+ parallel Steane patches fit comfortably with margin).
4. **Bivariate-bicycle code**: implement a smaller variant of IBM's `[[72,12,6]]` code on Heron-r2 and compare to Steane.

7 Reproducibility

All code, raw counts, and run records are at `github.com/Zynerji/systrophe`:

- Encoder + decoder + syndrome circuit: `experiments/steane_logical_qubit.py`
- Submission record: `experiments/results/steane_kingston_n4_submitted.json`
- Analysis JSON (logical rate, physical rates, sigma): `experiments/results/steane_kingston_n4_analysis.json`
- Press-release retraction: `launch/viral_press_release_156q_qec.md`

References

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